TEXORA · LECTURE SHEET

Knitted Fabric Structure: Introduction, Notation and Types

The second half of this course changes the forming principle: instead of two yarn systems interlacing, one yarn system is bent into loops and the loops are intermeshed. This sheet fixes the knitted vocabulary — loop anatomy, wales and courses, the notation systems, and the classification — in exactly the terms used by TE-3101 and TE-3102, so the two courses read as one.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 9 · LEVEL 3 TERM 2

1. From interlacing to intermeshing

A **knitted fabric** is formed by bending yarn into loops and drawing each new loop through the loop of the previous row. The elementary unit is the **stitch (loop)**, with three parts: the **needle loop** — the head arc plus its two legs, formed in the needle hook; the **sinker loop** — the arc joining two neighbouring needle loops at their feet, lying between the needles in weft knitting. Loops stand in columns and rows: a **wale** is a vertical column of loops made by one needle; a **course** is a horizontal row of loops made in one knitting cycle. Density is therefore counted as **wales per cm (WPC)** and **courses per cm (CPC)**, and their product is the **stitch density** (loops per cm²). The vertical column/row language replaces the warp/weft language of Topics 1–8 — never mix the two in one description.

The fabric has a **technical face** (the side where the V-shaped legs of the needle loops show) and a **technical back** (where the loop heads and sinker arcs show). Which side is "up" in a fabric is decided by the structure: in rib and interlock both faces show legs in some wales — the structures of Topic 11.

2. Notation — the three records, as in TE-3101

Knitted structures are recorded in the same three ways introduced in TE-3101/3102, and this course uses them unchanged:

- **Loop diagram:** the yarn path drawn as it lies — truthful, slow (the loop sketch of TE-3206/2 is one).
- **Needle (dot) diagram:** a grid, one dot per needle/wale per course; in this course's convention **● = a loop drawn to the face side at that wale, ○ = a loop drawn to the reverse side**; rows read as

courses, columns as wales. Tuck and miss are added as further symbols where a sheet needs them and are always defined on that sheet.

- **Cam / needle arrangement:** which needles are set to knit, tuck or miss at each feeder — the machine instruction matching the diagram.



Figure 1 — Needle (dot) notation in the TE-3101 convention: ● face loop, ○ reverse loop. Jersey draws every loop one way; rib alternates the facing wale by wale. Purl (Topic 12) will alternate it course by course — the same two symbols carry all four basic structures.

3. The three loop-forming actions

ELEMENT	NEEDLE ACTION	STRUCTURAL EFFECT
Knit loop	Needle clears fully, takes the yarn, knocks the old loop over	The normal stitch; the reference element
Tuck loop	Needle rises only to tuck height: new yarn held in the hook together with the old loop, both knocked over next time	Wider, thicker fabric; held loop lengthens later (cardigan, piqué, honeycomb knits)
Miss / float	Needle does not rise; the yarn floats behind it	Restrains widthways stretch; floats on the back (jacquard backs, inlay)

4. Classification of knitted structures

By formation: weft knitting — one yarn travels course by course across the fabric (jersey, rib, interlock, purl and their derivatives; the fabric can be unravelled course by course); **warp knitting** — every needle has its own yarn travelling lengthwise with underlaps (tricot, raschel families; cannot be unravelled course-wise). **By structure (weft):** the four basics of TE-3101 — **plain, rib, interlock, purl** — and the **derivatives** built from them by tuck, miss, transfer and yarn changes (piqué, lacoste, fleece, Milano rib, cardigans…). Topics 10–12 take the basics in turn as *structures* — their geometry, parameters and numbers: Topic 10 plain (single jersey), Topic 11 the double-knit pair rib and interlock, Topic 12 purl. **By use of needles:** single-bed (single jersey) versus double-bed structures.

5. The structural parameters (preview of Topics 10–12)

Five numbers specify any knitted structure well enough to reproduce it, and TE-3206/2 measures all five from a sample:

- **Stitch (loop) length** — yarn length in one loop, mm; the master variable of knitting.
- **WPC, CPC, stitch density** — the realised geometry in the relaxed state.
- **Tightness factor** K = √tex ÷ stitch length (mm) — compactness of the structure independent of yarn count (worked fully in Topic 10).
- **Yarn count** (tex or Ne) and **fabric weight** (GSM), tied together by GSM = WPC × CPC × stitch length × tex ÷ 100.
- **Relaxation state** — dry-relaxed, wet-relaxed or fully relaxed/finished: the same fabric gives different WPC/CPC in each, so the state is part of the specification.

Glossary

Stitch / loop — the unit of knitted structure: needle loop plus connecting sinker loop. **Wale** — column of loops from one needle. **Course** — row of loops from one knitting cycle. **Technical face / back** — sides showing loop legs / loop heads and sinker arcs. **Dot notation** — grid record of loop facing per wale and course. **Tuck** — yarn held in the hook without clearing. **Float (miss)** — yarn passing behind a needle unknitted.

Key points

• Knitting intermeshes loops; its grid is wales × courses, not ends × picks.

• One notation serves the whole course: ● face loop, ○ reverse loop, in dot diagrams whose rows are courses.

• Four basics (plain, rib, interlock, purl) + tuck/miss generate the weft-knitted family; five parameters (stitch length, WPC, CPC, count, GSM, with relaxation state) specify a fabric completely.

Practice

1. Write the dot notation for one course of 2×2 rib over 8 wales.
2. A fabric's loops can each be pulled out, row by row, from the last row knitted. Weft or warp knitted — and why?
3. State which of the four basic structures is identified by: facing alternating (a) by wale, (b) by course, (c) not alternating but doubled back-to-back.

4. Why must a knitted specification state the relaxation state when a woven specification needs no equivalent entry?

Answer key

1. ● ● ○ ○ ● ● ○ ○ (two face wales, two reverse wales, repeating).
2. Weft knitted: its single yarn runs course by course, so unravelling releases one course at a time from the last-knitted end. A warp knit cannot do this — every wale carries its own yarn locked by underlaps.
3. (a) Rib. (b) Purl. (c) Interlock — two ribs intermeshed, each wale backed by a loop of the other rib.
4. Knitted loops are held only by intermeshing and the bent yarn keeps trying to re-shape the fabric after knitting: WPC and CPC continue to change with relaxation and finishing, so a count without its state is ambiguous. A woven cloth's interlacing locks its thread spacings far more rigidly at the loom state.

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